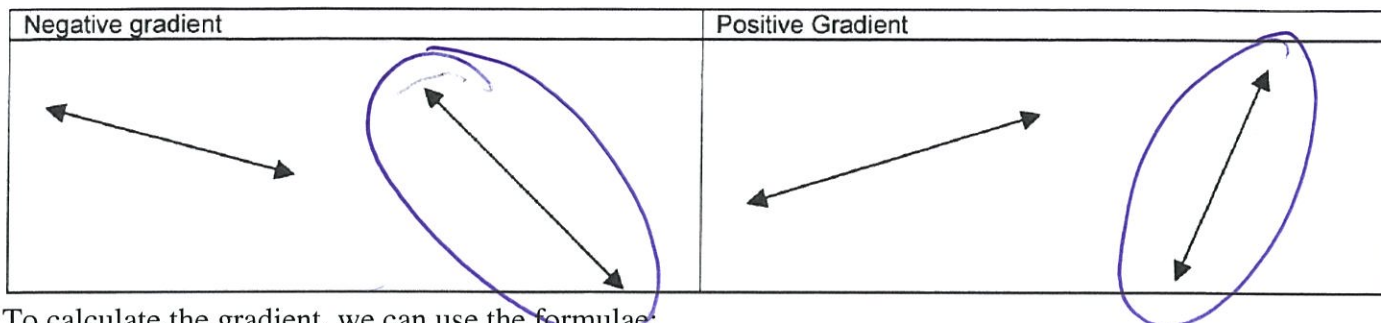


Gradient, y-intercept and the equation of a line

- Remember the **gradient** of a line is the slope of a line. The slope of a line is determined by how much we then go up and how far we go across.
- Gradient can be represented as a fraction, decimal or whole number.

Example

Circle the line that would have a larger value (number) for its gradient.



To calculate the gradient, we can use the formulae:

$$\text{gradient (m)} = \frac{\text{rise}}{\text{run}}$$

How to determine the gradient of the line

- Step One: Locate two points on the line (find ones that have two grid lines meeting)
- Step Two: Start from one point and then move vertically until you reach in line with the second point. Then move towards the second point counting how many places you have moved. (Up is positive and down is negative). This is the rise.
- Step Three: Start from one point and then move horizontally across (left is negative and right is positive) until you reach in line with the second point. Write the number of places you moved down. This is the run.
- Step Four: Simplify if necessary.

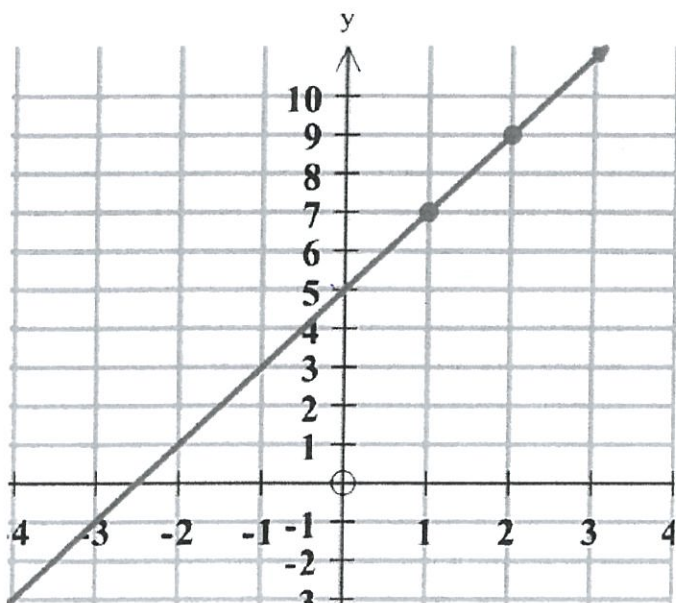
How to determine the y-intercept of the line

- Step One: Identify where the line crosses the y axis ($x=0$)
- Step Two: The y value is the y intercept.

How to determine the equation of the line

- Step One: Identify where the line crosses the y axis ($x=0$)
- Step Two: The y value is the y intercept.

Example:

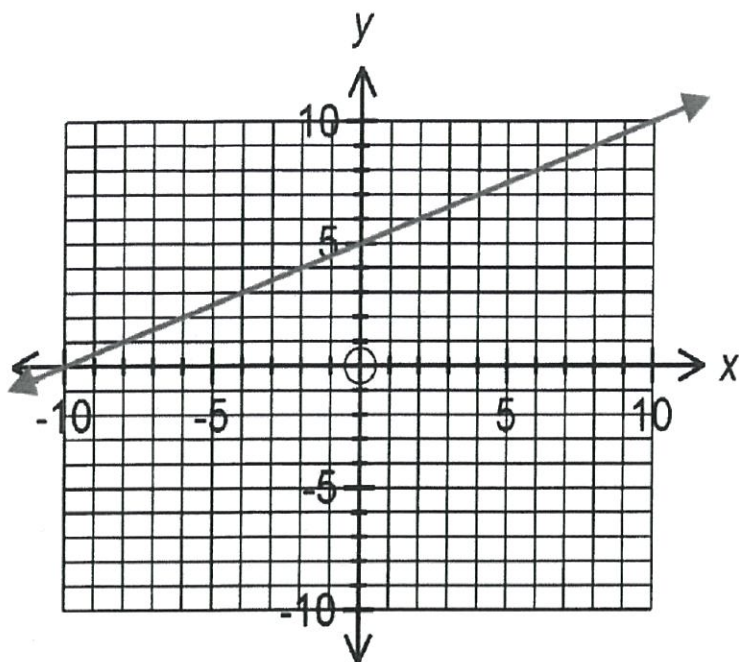


$$\text{gradient (m)} = \frac{\text{rise}}{\text{run}} = \frac{2}{1} = 2$$

$$\text{y-intercept (c)} = 5$$

Equation of the line $y = mx + c$

$$y = 2x + 5$$

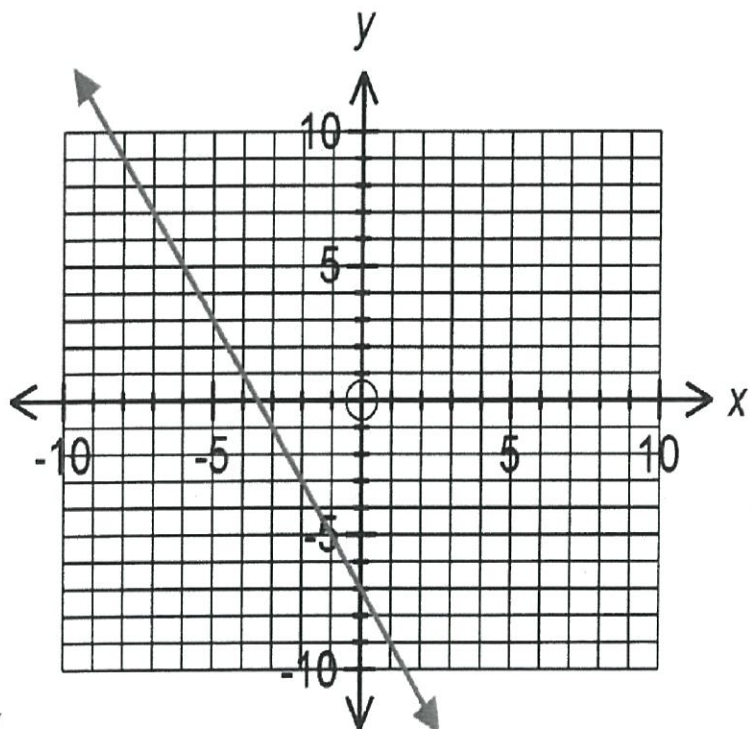


$$\text{gradient } (m) = \frac{\text{rise}}{\text{run}} = \frac{1}{2}$$

$$y - \text{intercept } (c) = 5$$

Equation of the line $y = mx + c$

$$\underline{y = \frac{1}{2}x + 5}$$

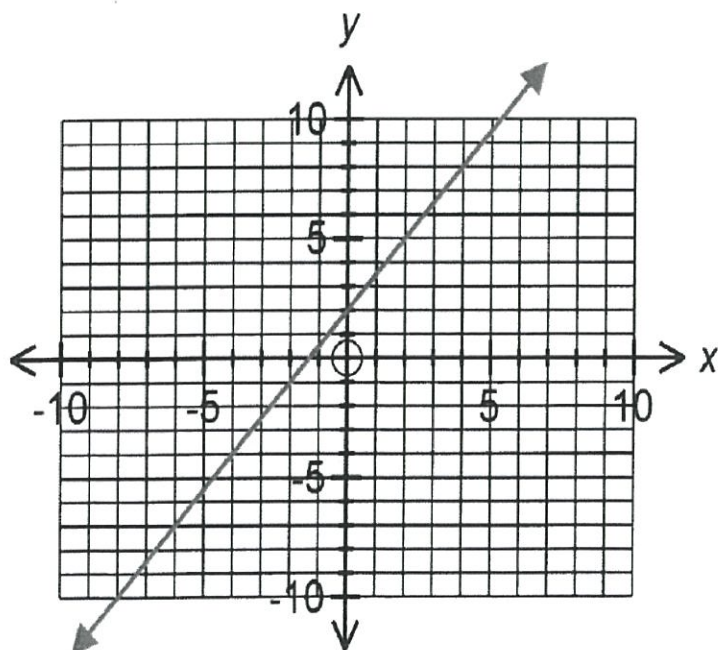


$$\text{gradient } (m) = \frac{\text{rise}}{\text{run}} = \frac{-2}{1} = -2$$

$$y - \text{intercept } (c) = -7$$

Equation of the line $y = mx + c$

$$\underline{y = -2x - 7}$$



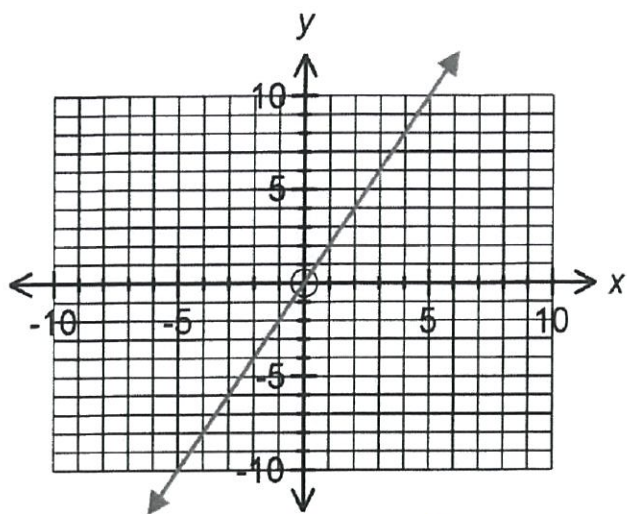
$$\text{gradient } (m) = \frac{\text{rise}}{\text{run}} = \frac{3}{2}$$

$$y - \text{intercept } (c) = 2$$

Equation of the line $y = mx + c$

$$\underline{y = \frac{3}{2}x + 2}$$

Determine the Gradient of the following Lines

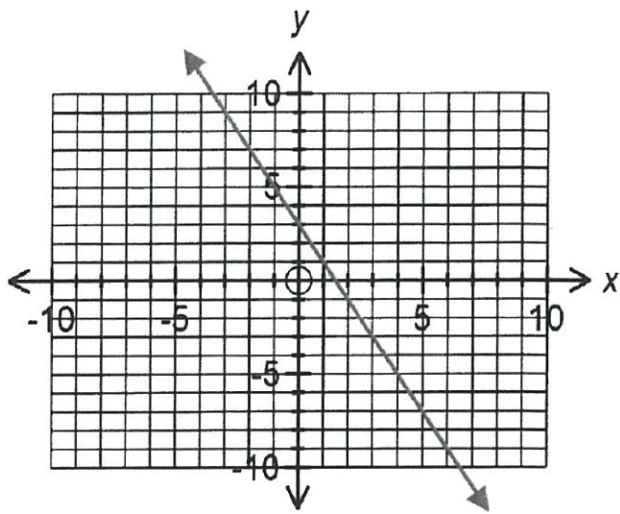


$$\text{gradient (m)} = \frac{\text{rise}}{\text{run}} = \frac{2}{1} = 2$$

$$y - \text{intercept (c)} = 0$$

Equation of the line $y = mx + c$

$$y = 2x$$

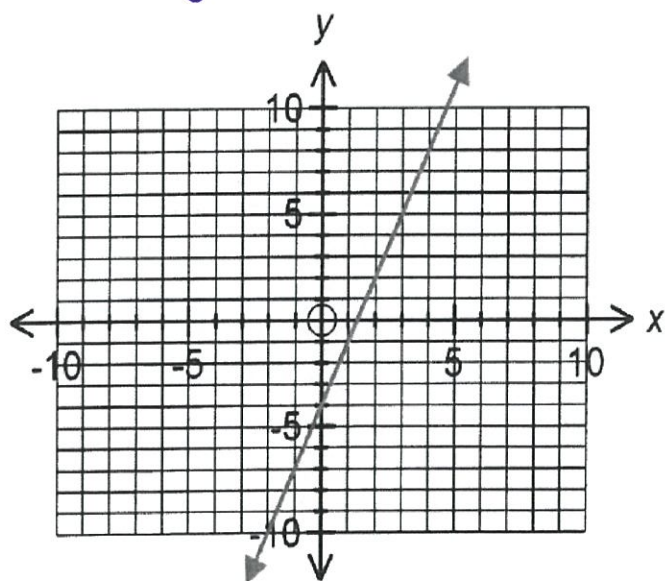


$$\text{gradient (m)} = \frac{\text{rise}}{\text{run}} = \frac{-2}{1} = -2$$

$$y - \text{intercept (c)} = 3$$

Equation of the line $y = mx + c$

$$y = -2x + 3$$

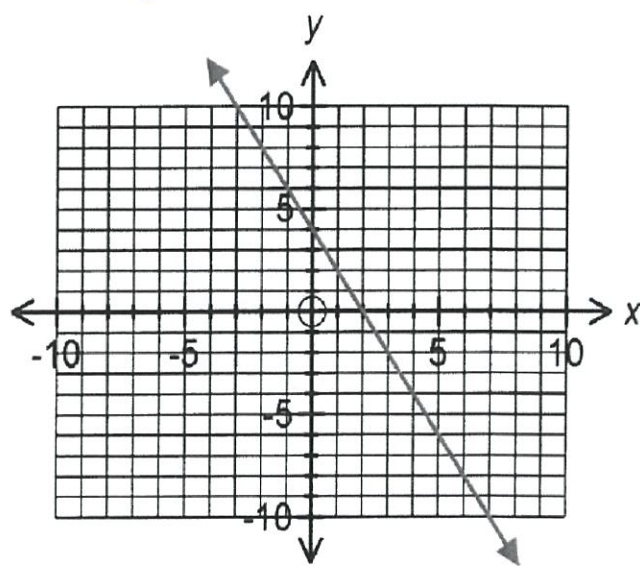


$$\text{gradient (m)} = \frac{\text{rise}}{\text{run}} = \frac{3}{1} = 3$$

$$y - \text{intercept (c)} = -4$$

Equation of the line $y = mx + c$

$$y = 3x - 4$$

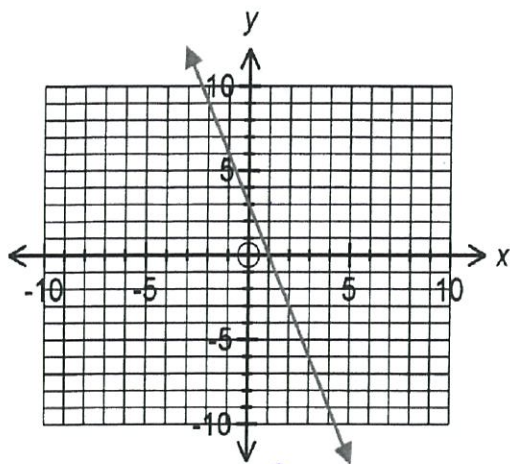


$$\text{gradient (m)} = \frac{\text{rise}}{\text{run}} = \frac{-2}{1} = -2$$

$$y - \text{intercept (c)} = 4$$

Equation of the line $y = mx + c$

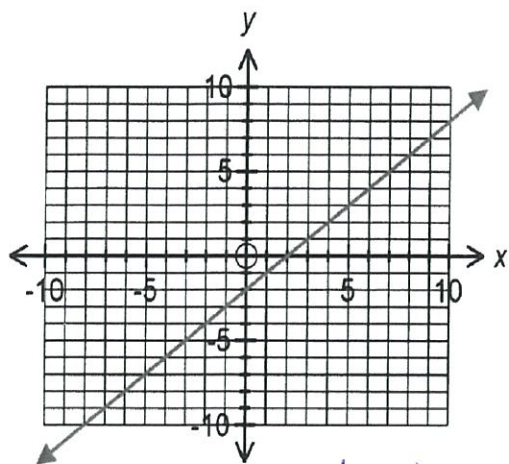
$$y = -2x + 4$$



gradient = $-\frac{3}{1} = -3$

y - intercept = 3

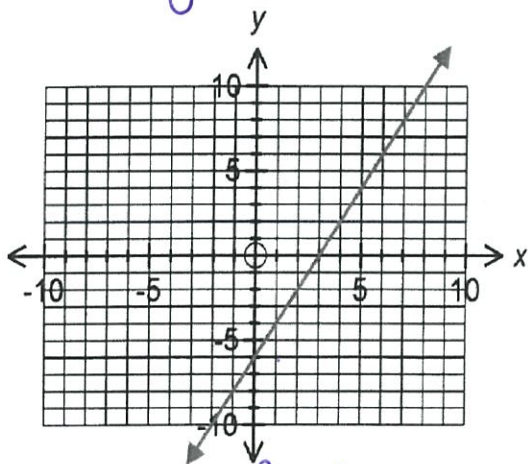
Rule: $y = -3x + 3$



gradient = $\frac{1}{1} = 1$

y - intercept = -2

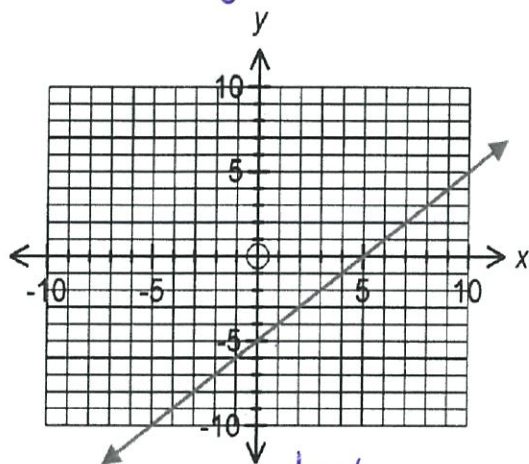
Rule: $y = x - 2$



gradient = $\frac{2}{1} = 2$

y - intercept = -6

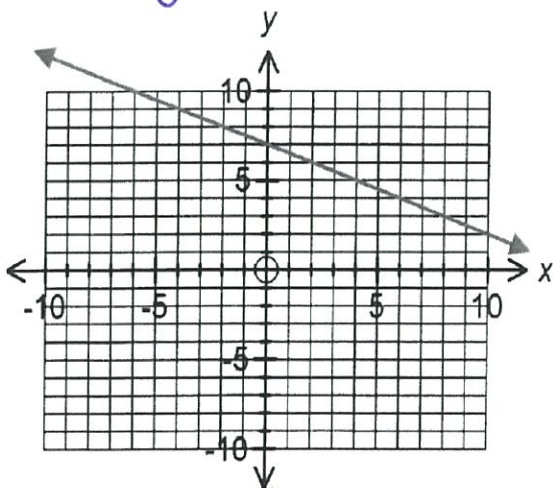
Rule: $y = 2x - 6$



gradient = $\frac{1}{1} = 1$

y - intercept = -5

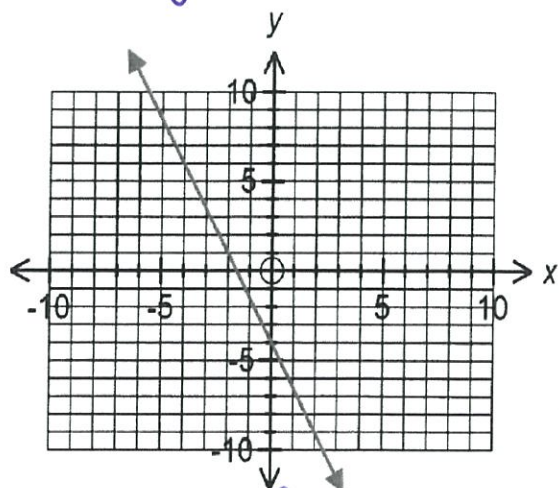
Rule: $y = x - 5$



gradient = $-\frac{1}{2}$

y - intercept = 7

Rule: $y = -\frac{1}{2}x + 7$



gradient = $-\frac{2}{1} = -2$

y - intercept = -4

Rule: $y = -2x - 4$